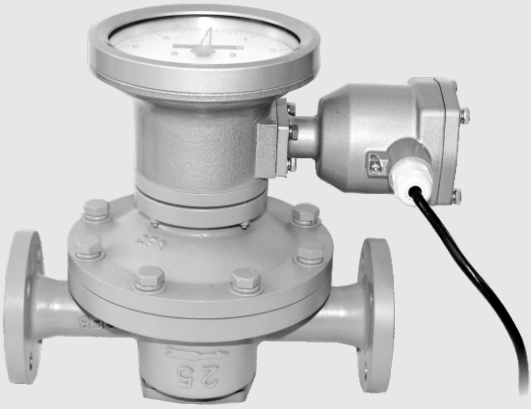


IOTAFLOW
SYSTEMS PVT. LTD

OVAL GEAR FLOW METER

MANUAL AND INSTRUCTIONS

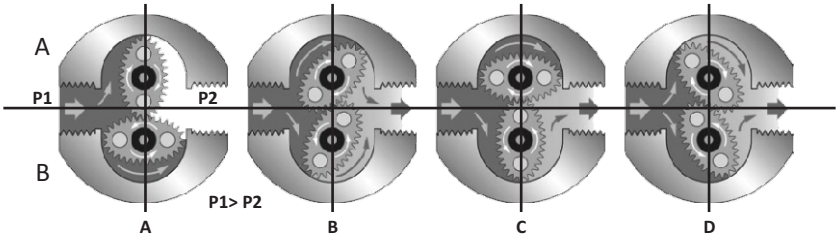
" If You Want To Manage It, Measure It "

www.iotaflow.com

1. Summarize

Oval wheel flow meter is used for pipeline liquid flow measurement of continuous or intermittent high-precision measurement instrumentation. It has high accuracy, good reliability, light weight, long life, easy installation etc. It is a typical positive displacement flow meter product. Oval Gear Flow meter measures mainly by the measurement cavity, sealing the composition of the coupling and counters. The measurement can be cumulative and instantaneous flow of liquid. Coupled with intelligent control device and flow meter, can get remote measurement and control. It widely used in petroleum, chemical, electric power, metallurgy, transportation, food processing, medicine, defense, commerce and trade sectors such as oil and petroleum products, chemical solutions and other fluids accurate measurement.

2. Working principle

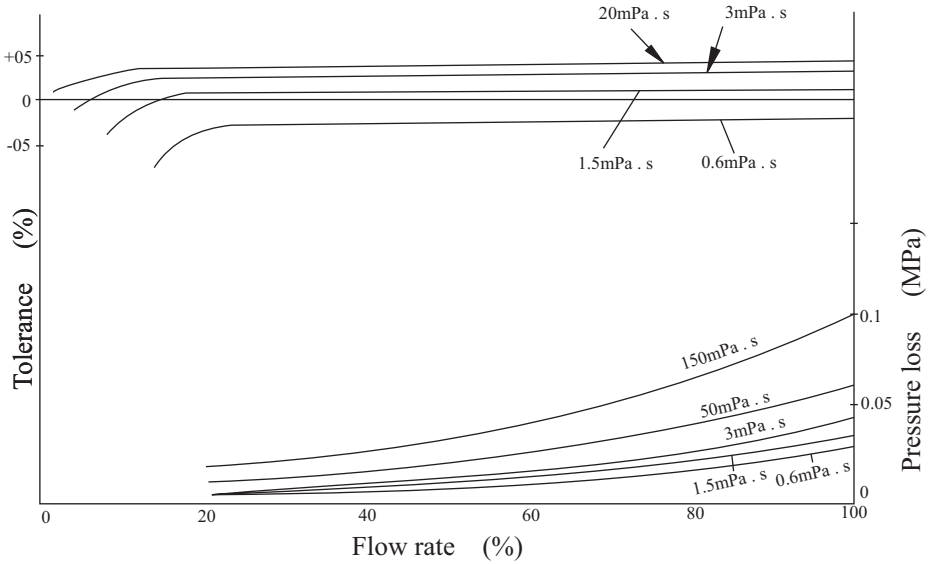


When the measured liquid flowing through the measuring chamber when the pressure in the formation of the inlet and outlet meter, waist-round under the impetus of pressure in the second rotation. At the same time fixed on the waist by a pair of drive axle on the gear, there are two waist-round to maintain a continuous rotation. With the waist wheel rotation, the liquid is continuously discharged through the metering chamber flow meter. Convection over the liquid for each volume is four times the volume of metering chamber. Through the sealing coupling, deceleration institution, after slowing down the rotation number passed to the counter, counters that the instantaneous direction of liquid flow and cumulative flow.

3. Main parts and materials combinations

Materials / Parts	AL			EL						LL
	Cast Iron	Aluminum	Stainless Steel	Cast Steel	Aluminum	Stainless Steel	Iron	Aluminum	Iron	Stainless Steel
Body	☆			☆						☆
Oval Gear		☆			☆					☆
cover	☆						☆			☆
shaft			☆			☆				☆
Bearing			☆			☆				☆

4. Performance and pressure drop curves

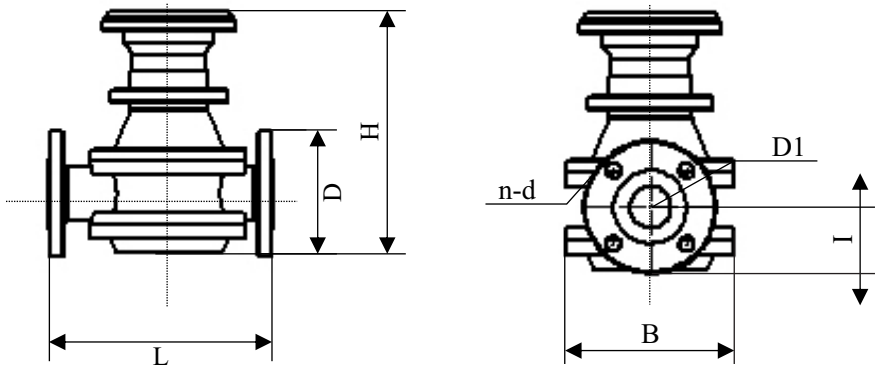


5. Spcifications

1. Working pressure (MPa) : 0.6, 1.0, 1.6, 2.5, 4.0
2. Temperature (°C) : -10~60
3. Accuracy grade : 0.5, 0.2
4. Flow Range (m /h)

Diameter(mm)	Liquid viscosity 3.0~150mpa.s	Liquid viscosity 3.0~150mpa.s	Liquid viscosity 2~200mpa.s	Liquid viscosity 3.0~150mpa.s	Liquid viscosity 3.0~150mpa.s	Liquid viscosity 2~200mpa.s
15	0.25~2.5	0.5~2.5	0.3~1.5	0.5~2.5	0.83~2.5	0.3~1.5
20	0.25~2.5	0.25~2.5	0.6~3.0	0.5~2.5	0.83~2.5	0.6~3.0
25	0.6~6.0	1.2~6.0	1.2~6.0	1.2~6.0	2.0~6.0	1.2~6.0
40	1.6~16	3.2~16	3.0~15	3.2~16	5.3~16	3.0~15
50	2.5~25	5.0~25	4.8~24	5.0~25	8.3~25	4.8~24
80	6.0~60	12~60	12~60	12~60	20~60	12~60
80 II	12~100	22~100	-	20~100	33.3~100	-
100	10~100	20~100	20~100	20~100	33.3~100	20~100
150	40~200	40~200	38~190	100~200	66.7~200	38~190
Accuracy grade	0.5			0.2		

6. Overall and installation dimensions



Oval Gear Flow meter Dimensions

Model	L	H	I	B	D	D1	n-d
15 mm	160	238	52	100*100	G11/4		
20 mm	160	238	52	100*100	90	65	
25 mm	180	260	68	120*120	G1/		
40 mm	240	290	88	170	150	110	4-17.5
50 mm	265	305	97	190	165	125	4-C17.5
80 mm	265	340	122	246	200	160	8-17.5
100 mm	515	425	164	305	220	180	8-18
150 mm	560	550	225	455	285	240	8-22

7. Oval wheel flow meter installation

(1) Flow meter should be installed after the filter, the two arrows pointing in the body on the line with the flow direction
(2) If there is gas in the pipeline, flow meter should be installed after the gas separator
(3) Regardless of pipe is installed vertically or horizontally, the waist bearings of flow meter installed into a horizontal position axle (the counter should be perpendicular to the ground)
(4) Flow meter installed in the right circumstances, and if difficult to see clearly, can be rotated 180 degrees to the counter, or 90 degrees
(5) Throttle valve should be installed in the meter inlet, opening and closing valve installed in the exit of the use of open-close valve slowly to start, do not suddenly open valve
(6) Prohibited the use of scan-line steam through the flow meter
(7) In the continuous use sector, need to add bypass pipe flow meter
(8) Pre-installation flow meter, piping required to wash, rinse used straight pipe section (alternative flow meter position) to prevent the welding slag and debris from entering flow meter
(9) Check water is strictly prohibited cast iron, cast steel material composition of flow meter
(10) When using the flow meter size should not exceed the technical requirements. Flow meter to work at maximum flow rate 70 ~ 80% is good.

8. Flow range (Accuracy is 0.5%)

Unit: m /h

Diameter	Oil (mpa.s)					
	Heavy oil 2.0~150	Light oil 0.6.~2.0	Heavy oil 2.0~150	Light oil 0.6.~2.0	Heavy oil 2.0~150	Light oil 0.6.~2.0
25	0.6~6	1.2~6	1.2~6	1.5~6	1.5~6	2~6
40	1.6~16	3.2~16	3.2~16	4~16	4~16	5~16
50	2.5~25	5~25	5~25	6~25	6~25	8~25
80	6~60	12~60	12~60	15~60	15~60	20~60
100	10~100	20~100	20~100	25~100	25~100	33~100
150	25~250	50~250	50~250	60~250	60~250	80~250
Accuracy	0.5		0.3		0.2	

9. Common cause of the malfunction, remove method

Failure	Reason	Measures	Notes
Roots can not divert	1. There is debris in the pipeline 2. The tested liquid containing more then debris 3. Filter damaged, debris into the table, waist wheel stuck Not turn	Take apart and wash instrumentation and piping, repair filters, repair of Oval wheel surface Nut turn	
Axial seal leakage coupling	Filer wear and tear or lack of seal oil	Cover or replace the fill screw pressed, plus seal oil	
Pointer rotation unstable or stop-and-go	Pointer, screw, waist wheel, moving parts are loose or not flexible rotation	Replacement of bearing, repair of varying tooth Department metering box wall and the gears, enabling rotation and flex	After repair to re-verify
The error is too large, but the difference between the minimum maximum error of less then 1%	After the period of greater then or repair using the changes in clearance	Re-calibration adjustment	For the 0.2 meter maximum difference between the minimum error does not exceed ± 0.17%
No signal of Letter device	1) the poor location writing block	To re-adjust the position so, before and after the move	
No signal of Letter device	1) the poor location writing block	To re-adjust the position so, before and after the move	
	2) reverse polarity	To re-change the + then the red line - then the black line	
There is leakage sealing parts	Aging O-ring failure	Replace the O-ring seal grease plus Tu	

Notes

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Contact Us:-

IOTAFLOW SYSTEMS PVT. LTD.

A-2/13, Mayapuri Industrial Area
Phase-II, New Delhi - 110064
Call: +91 9910693866
+91 7982674054
Email: contact@iotaflow.com

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